

Reducing image recognition accuracy loss from radiation in analog memristive chips

Final Report

Group #38

**Hoyeol Bae
Wataru Tamaki
Benjamin Tung
Zachary White
Justin Weidmann**

**Mentor:
Dr. Matthew Marinella**

EEE 489 - Spring 2025
April 30th, 2025

Table of Contents

Table of Contents	1
List of Illustrations	2
1. EXECUTIVE SUMMARY	3
2. INTRODUCTION	4
- Problem Statement	4
- Project Objectives	4
- Roadmap	4
3. SCOPE OF PROJECT	5
- Conductance Drift Due to Radiation	5
- Conductance Drift Due to Time and Temperature	6
- Target Specifications	7
4. PLANNING/SET-UP	8
5. TECHNICAL WORK COMPLETED	10
- Alpha Correction Factor	10
- Training with Noise	11
- Training with Noise and Alpha in the Loop	12
6. FUTURE WORK	13
7. ABET CRITERION	13
- Sustainability	13
- Manufacturability	13
- Political	13
8. CONCLUSION	14
References	15

List of Illustrations

Figure 1: Artistic rendition of gamma rays incident on the surface of a memristive chip	5
Figure 2: Measured conductance of the memristive cells of TID	6
Figure 3: Diagram of individual SONOS cell	6
Figure 4: Global Drift Compensation	8
Figure 5: Training with Noise Diagram	9
Figure 6: Training with Noise Backend	9
Figure 7: Left: Before Correction; Right: After Correction	10
Figure 8: TID vs Accuracy, using Alpha Correction Factor	11
Figure 9: Standard Deviation and Percent Error of Dot Products	11
Figure 10: TID vs Accuracy, Training with Noise	12
Figure 11: Alpha in the Loop Accuracy	12
Table 1: Model 5% Roll-Off Point	13

1. EXECUTIVE SUMMARY

In a world increasingly reliant on artificial intelligence, the demand for low-power, high-efficiency computational hardware has never been greater. Memristive chips, with their ability to perform calculations directly within memory, offer a promising solution to this challenge. These chips are especially valuable in applications like self-driving cars, where real-time image processing is critical but energy-intensive. However, their potential is limited by significant obstacles: conductance drift and voltage spikes, often caused by radiation and temperature fluctuations, which undermine their accuracy and reliability [1].

Our project focuses on solving these problems by introducing two key innovations: the alpha correction factor and noise injection during training. The correction factor adjusts for radiation-induced drift, while noise injection helps models adapt to interference. These techniques aim to stabilize the chip's performance under challenging conditions, such as high-radiation environments in space. Together, they significantly improved the chip's resilience-maintaining accuracy up to 57.3 krad of radiation exposure, a major improvement over the baseline of 4.17krad. All testing was conducted using CrossSim, a simulation platform, with a ResNet 32 image recognition model [1].

The work was conducted in two stages. In the first semester, the focus is on understanding the technology, running simulations using CrossSim, and testing ideas virtually. The second semester focused on transitioning to physical hardware and preparing for real-world testing, such as radiation exposure in ASU's lab environment. We completed the project within a modest academic budget and used commercially available SONOS chips, ensuring scalability and manufacturability.

The impact of this work extends beyond technical success. By making AI hardware more energy-efficient and robust in harsh environments, this project supports advances in sustainability, autonomous vehicles, military systems, and long-term space missions.

2. INTRODUCTION

Problem Statement: This project focuses on improving memristive chips for analog image recognition. These chips are crucial for low-power matrix multiplication, especially in applications like self-driving cars, where real time image processing is needed but can be power-intensive. Memristive chips offer up to four times more power efficiency compared to traditional digital chips [1].

However, key challenges arise from conductance drift and voltage spikes caused by radiation and temperature variations, leading to reduced image recognition accuracy. These effects limit the reliability of memristive chips in harsh environments, such as space or high-radiation zones, where AI hardware must remain accurate over time [1].

Project Objectives: The growing demand for energy-efficient real-time image processing, especially in applications like autonomous vehicles and space systems, has increased interest in analog memristive chips. These chips consume significantly less power than traditional digital ones but are vulnerable to performance loss in high-radiation environments. This project aims to improve their radiation tolerance so that classification accuracy remains within $\pm 10\%$ of its baseline after exposure to up to 50 krad. Achieving this goal would make memristive chips a reliable and efficient option for deployment in space, military, and other energy-constrained systems.

Roadmap: The first section, *Executive Summary*, introduces memristive chips and highlights their potential to enable low-power, high-efficiency AI computations in extreme environments such as space exploration and autonomous systems. The second section, *Introduction*, outlines the project's motivation, objectives, and significance, emphasizing the need for resilient AI hardware that maintains performance under radiation exposure. The third section, *Scope of Project*, defines the technical limitations addressed by the project, focusing on conductance drift caused by radiation and establishing performance benchmarks for success. The fourth section, *Planning/Set-up*, describes the early project phases, including simulation environment preparation, radiation modeling using CrossSim, and strategy selection. The fifth section, *Technical Work Completed*, presents the implementation and results of the alpha correction factor, noise injection during training, and the combined alpha-in-the-loop approach. The sixth section, *Future Work*, discusses the planned transition from simulation to physical validation through irradiation testing. The seventh section, *ABET Criterion*, reflects on the project's sustainability, manufacturability, and broader societal and political relevance. The final section, *Conclusion*, summarizes the project's contributions and its significance in advancing analog AI hardware for use in radiation-prone environments.

3. SCOPE OF PROJECT

There are several problems that memristive chips currently suffer from, but only a few of these problems will be addressed by this project. These issues are voltage spikes due to radiation, conductance drift due to radiation, and conductance drift due to time. The scope of this project previously included all three of these issues, but the focus has been narrowed down to only the conductance drift due to radiation. Overcoming this obstacle would result in the most valuable benefit to the industry as a whole, since it would make memristive chips viable for space applications.

Conductance Drift Due to Radiation: When these memristive chips are deployed in the field (on a self-driving car, for example) there will be ambient radiation due to several sources. The sun is a notorious emitter of gamma rays, which are more than capable of penetrating through a vehicle's frame, and into the memristive chip's sensitive analog architecture. The most common source of background radiation is due to Radon, a gas that arises from the decay of naturally occurring Uranium in the ground, which emits alpha, beta, and gamma particles.

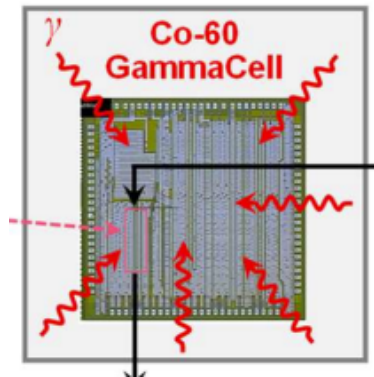


Figure 1: Artistic rendition of gamma rays incident on the surface of a memristive chip [3].

All forms of ionizing radiation are capable of interfering with the stability of the memristive cells, which require a precisely controlled conductance to gain accurate matrix multiply results.

These conductance values, which represent weights, are controlled via trapped charges. This radiation, being ionizing, is capable of physically stripping charges from their cells, thus altering the conductance of that cell and degrading the results of the matrix multiply.

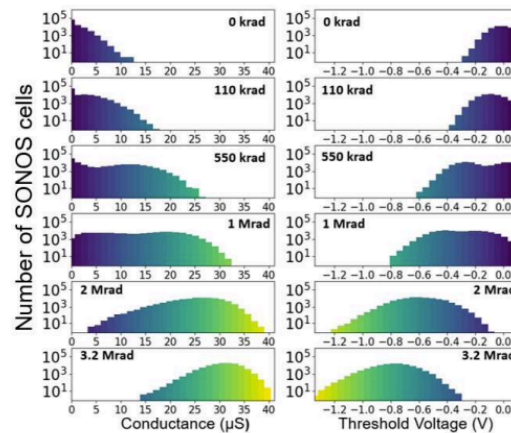


Fig. 2: (Left) Distributions of SONOS cell conductances in the array at increasing TID levels, (right) distributions of corresponding V_T values.

Figure 2: Measured conductance of the memristive cells when exposed to increasing doses of radiation (TID = Total Ionizing Dose) [4].

For space applications, these memristive chips are highly vulnerable to the large doses of radiation that are present, in the range of 0.1 to 10 Mrads/s, which can accumulate to the order of 100 krad over the lifetime of the spacecraft [2]. However, the memristive chips being used for terrestrial applications experience little conductance drift due to radiation, despite the presence of the sun and Radon gas. The effects of time and temperature on the conductance of the cells are much more significant in the performance of memristive chips being used on Earth.

Conductance Drift Due to Time and Temperature: The SONOS cells fall victim to electron leakage, just like any other circuit would. This is a relevant issue for the field in general, but will not be focused on for the scope of this project for a simple reason: In order to test the loss in accuracy over long periods of time, e.g. the lifetime of a self-driving car (~10-15 years) or the lifetime of a spacecraft (decades or more), preliminary experiments to measure the magnitude of accuracy loss over these time periods would take several years at minimum. This project is meant to span over the course of two semesters, so it is simply not feasible to collect this data in such a short period of time.

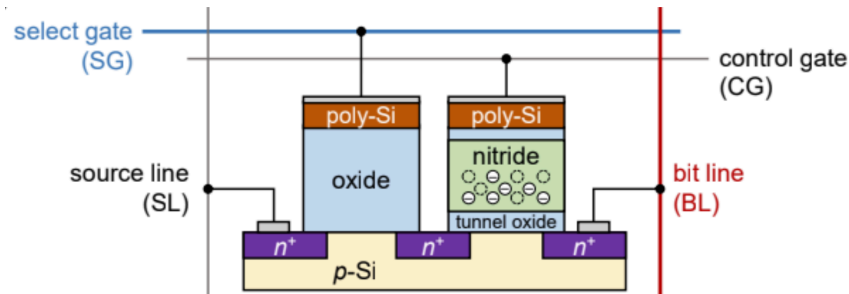


Figure 3: Diagram of an individual SONOS cell [4]. Note that this cell does not explicitly depict conductance drift due to time and temperature, but is a helpful image for visualizing what that drift would look like.

The number of electrons trapped in the nitride layer give rise to the programmed conductance values, and the precise charge that is stored in each cell is very important for the accuracy of the whole chip. These electrons can be “knocked loose” from the nitride layer via leakage due to imperfections in the oxide layer. Temperature will accelerate this process, due to the fact that the average kinetic energy of the electrons is higher at higher temperatures (by definition), and therefore have a higher tendency to escape from their energy bands and out of the nitride layer [3].

Target Specifications: The accuracy of the image classification algorithm can drop drastically from around 60% down to near 0% at Total Ionizing Doses of 2,000 rad or higher. (see Figure 8). In order for this project to be considered a success, the accuracy should maintain +/- 5% of its baseline value over the lifetime of the product the chip is implemented in. Note that, for the scope of this project, the “baseline accuracy” is defined as the accuracy of the chip before significant exposure to radiation. In practice, this would be measuring the accuracy immediately after manufacturing, or measuring the accuracy right before installing the chip onto a spacecraft, for example.

i) For use in self-driving car: The average lifespan of a terrestrial vehicle is around 10-15 years, and the yearly exposure to gamma rays on Earth is between 1 and 2 mSv (only gamma rays are considered, since they pose the greatest source of TID to memristive chips due to their high penetration power and abundance from outer space). For gamma rays, the unit of Sieverts has a 1:1 ratio with the unit of Gray, which has a 100:1 ratio with the unit of rads. Converting 1 mSv to rads, the resulting yearly dose of gamma rays on Earth is around 1 to 2 rads, which over the course of 10 years gives only 10 to 20 rads, completely insignificant to the performance of the chip, and would have nearly no effect on the accuracy.

ii) For use in space applications: As mentioned earlier, a spacecraft can be exposed to anywhere between 0.1 to 10 mrads per second, a massive increase to the dose that terrestrial vehicles receive. This buildup over time can result in radiation doses in the order of 10^5 [2]. This means that the memristive chip, with no protection to account for the resulting conductance drift, will see a drop to 0% accuracy in less than 10 years, which is significantly shorter than the lifespan of most spacecraft. For example, the International Space Station has been in orbit for 26 years, much longer than the current lifespan of memristive chips. As seen in Figure 8, the chip would see a 0% image classification accuracy at only around 100 krad, which does not satisfy the desired +/- 5% accuracy change over the lifetime of the spacecraft. Since the lifetime of a spacecraft is typically arbitrary, and varies depending on its purpose, this project defines “the lifetime of a spacecraft” to be 20 years. With this definition, the project will be considered a resounding success worthy of publication if the memristive chip can maintain +/- 10% accuracy from baseline after a TID of 100 krad. This achievement would represent leaps and bounds of progress in the field of analog computing, but it is a little unrealistic. A more realistic goal, considering the timeline and equipment available, would be to maintain a +/- 10% accuracy from baseline after a TID of 50 krad.

4. PLANNING/SET-UP

One of the first major keys to the project is understanding how the SONOS cell responds to the effects of radiation, and being able to simulate that using CrossSim, a Python based module that can replicate the properties of the memristive chips. Previous experimenters irradiated a SONOS cell in a radiation chamber up to 3.2 Mrad, and they collected information about conductance drift and voltage current characteristics that were implemented into a custom CrossSim chip model [3]. Using this chip model, it is possible to then simulate how the chip can perform dot products across a range of TIDs.

The next major challenge was figuring out how to properly correct for the drift caused by radiation. Another team from IBM had successfully figured out two techniques - applying a global drift compensation factor and training with noise [1], that successfully fixed drift caused by time. Due to the similarities in time based drift and radiation based drift, the team decided to apply those techniques to this experiment.

The first major technique to be implemented was the global drift compensation factor (also called the alpha correction factor). This correction factor scales the matrix-vector multiplication (MVM) by some value alpha, which is calculated from implementing MVMs at 0 TID, and MVMs at the TID the SONOS board is operating at, and dividing them:

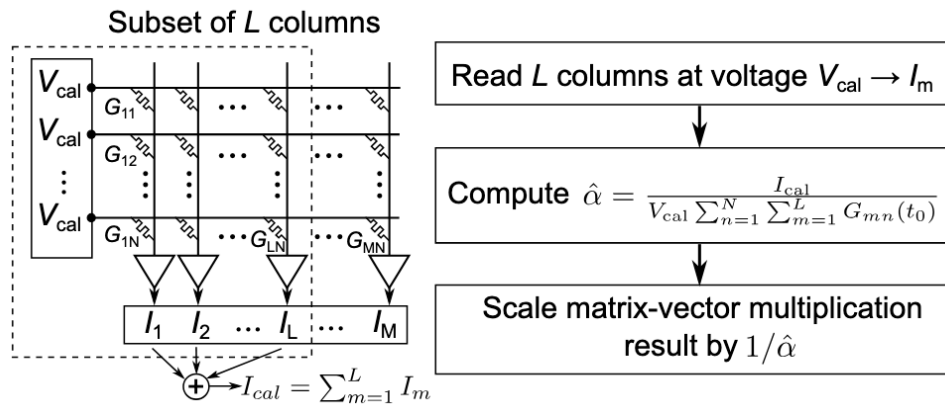


Figure 4: Global Drift Compensation [1]

The idea behind this technique is to fix the conductance drift caused by radiation by scaling the radiated matrix by the unperturbed matrix. Some major challenges to solve with this method will be how to apply this to a full machine learning network, as only doing the correction on one layer as shown above will not be enough to prove that the methodology works.

The second major technique is training the model with noise (the errors induced by radiation).

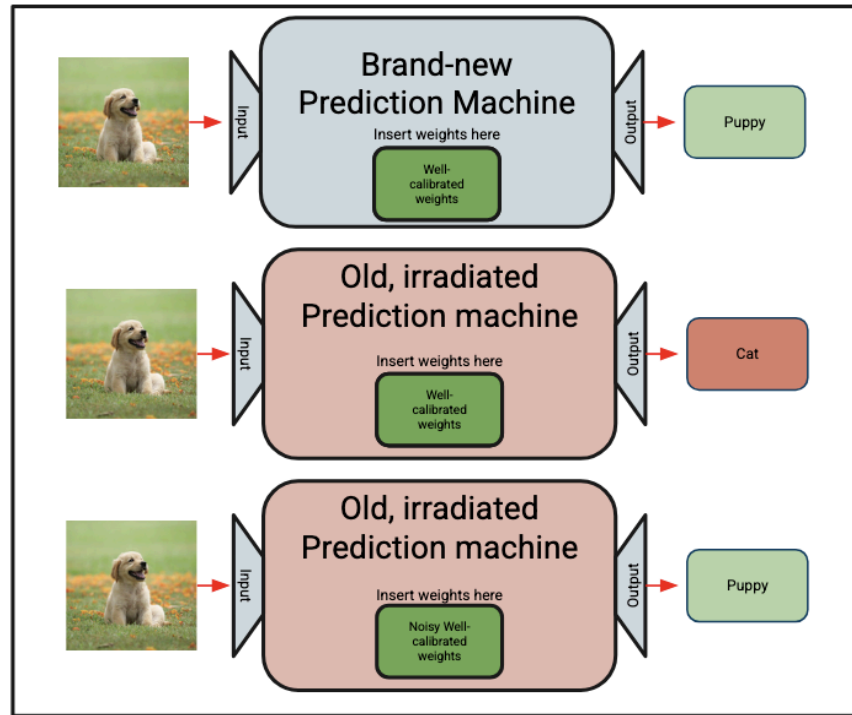


Figure 5: Training with Noise Diagram

The idea behind training with noise is that usually, if we train the model at 0 TID, that at higher TID doses it will be unable to properly perform inference. If we were to instead train the model at higher TIDs, it should be able to then properly do image recognition at those higher TIDs. The following training model was implemented in the code:

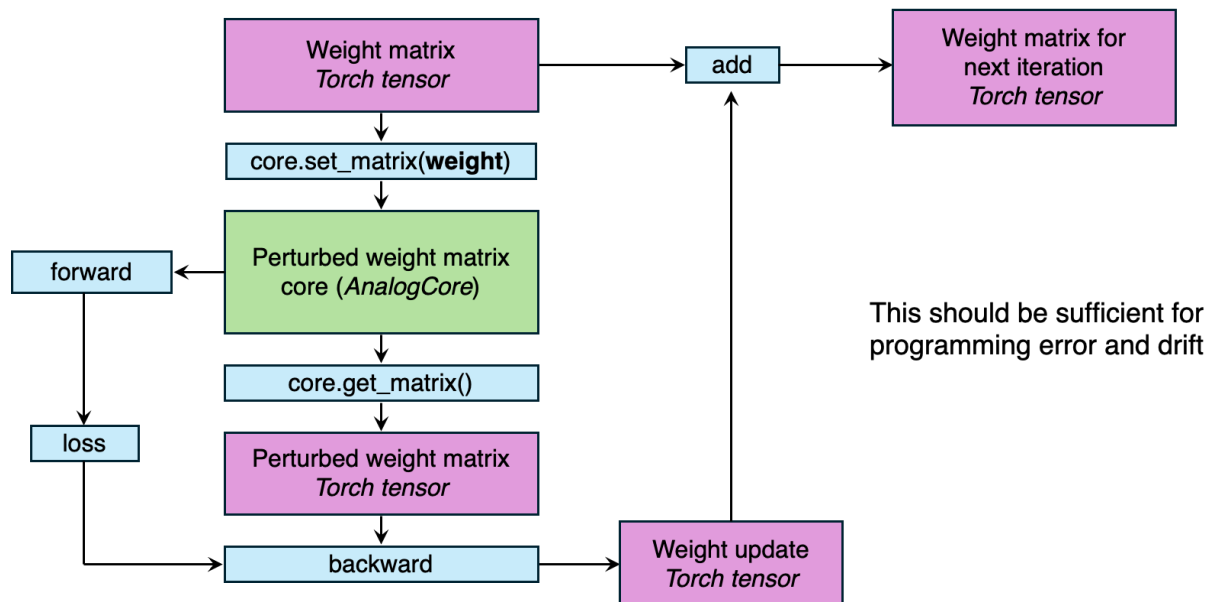


Figure 6: Training with Noise Backend

The weight matrix is injected with noise before being sent through the forward pass before being combined with the perturbed weight matrix torch tensor to go through the backward pass. Then, the weights are updated for the next iteration.

A major challenge will then be to figure out how to combine the alpha correction factor with training. By training the model with alpha in the loop, an extremely resilient model should be created.

5. TECHNICAL WORK COMPLETED

Alpha Correction Factor

The team was able to implement an alpha correction factor to correct for the conductance drift due to radiation in CrossSim for the entire machine learning network. The results demonstrate a marked improvement in accuracy upon using this method. The dot product calculations during the MVMs demonstrate a large decrease in percent error when doing this method on a single layer on the machine learning network:

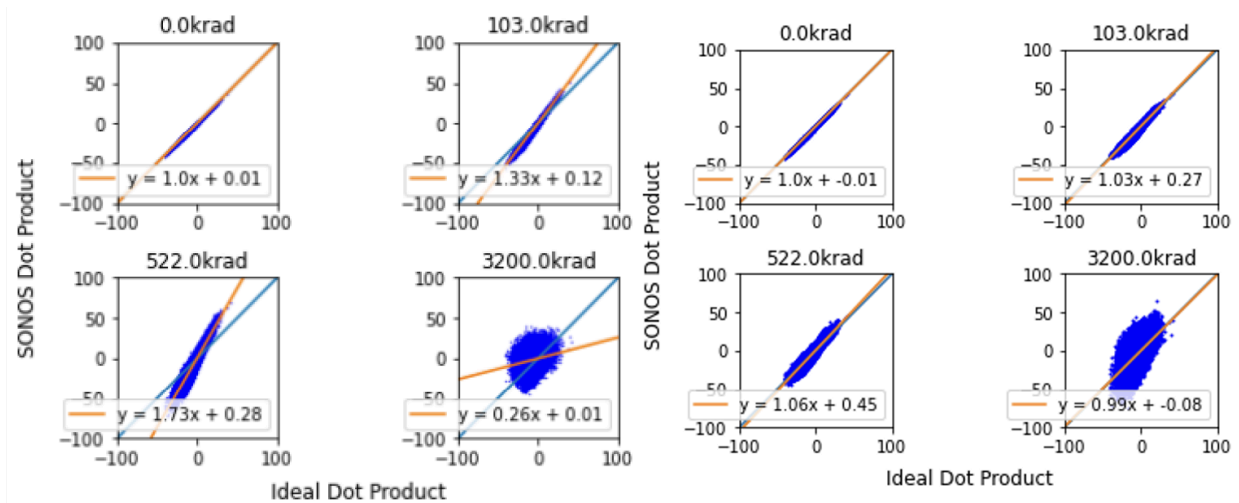


Figure 7: Left: Before Correction; Right: After Correction

After demonstrating that the alpha correction factor works on a single layer, a ResNet32 pytorch model was modeled and trained 20 times - with the best model boasting a peak test set accuracy of 56%. The correction factor was then implemented on the full network by calculating an alpha value for all 32 layers in the model before performing inference.

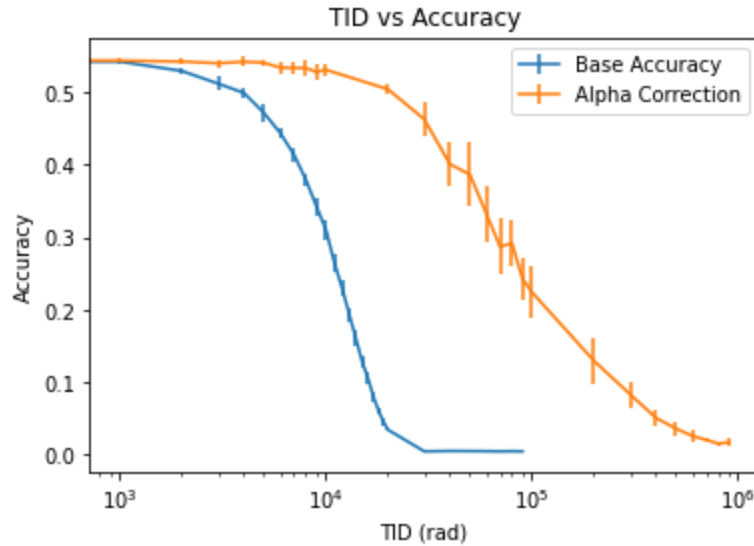


Figure 8: TID vs Accuracy, using Alpha Correction Factor

As seen above, using the alpha correction factor for the entire layer worked as expected, demonstrating a large accuracy improvement over not using alpha. However, the model should still be able to be further improved, as evidenced by the figure below.

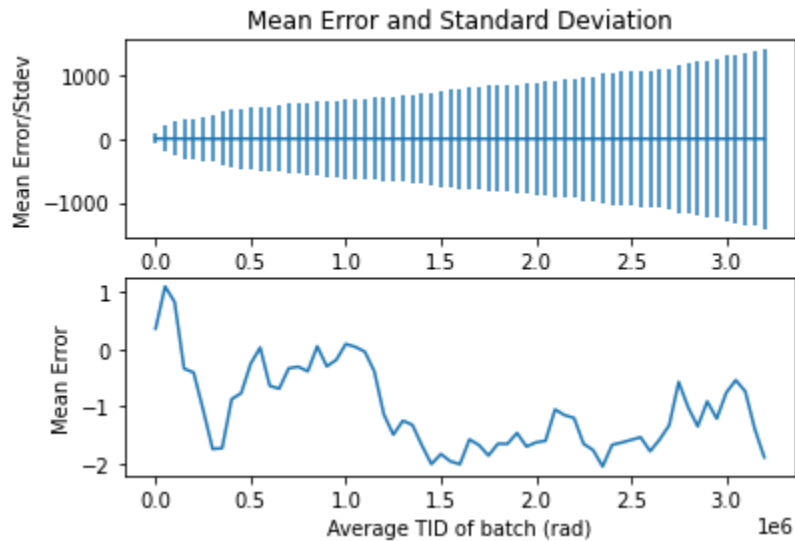


Figure 9: Standard Deviation and Percent Error of Dot Products

This figure demonstrates that although global drift compensation can more or less correct for all of the drift error, as seen by the mean error extremely close to zero, there is still a lot of variability in the error as seen by the extremely large standard deviation. This introduces the next error correction technique, training with noise, that should ideally be able to correct for the remainder of the error.

Training with Noise

By implementing the training diagram in Figure 6 in CrossSim, the following results were produced:

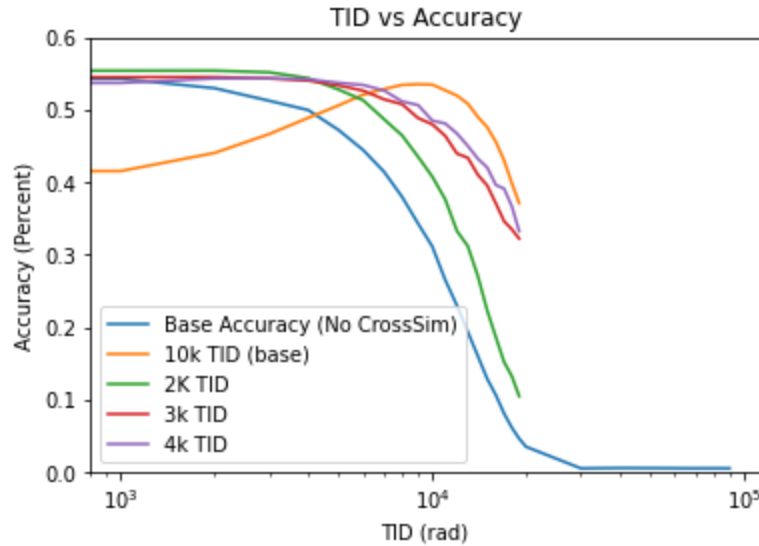


Figure 10: TID vs Accuracy, Training with Noise.

Training at lower TID levels (2k TID - 4k TID) resulted in a much more resilient model that remains accurate at 0 rad and maintains its accuracy further past the roll off of the base accuracy.. However, training at 10k TID resulted in poor results at 0 TID, which means that the 10k model most likely will not be reliable for general use.

Training with Noise and Alpha in the Loop

The last major step in the project was to combine the two previous techniques by calculating an alpha correction factor during training.

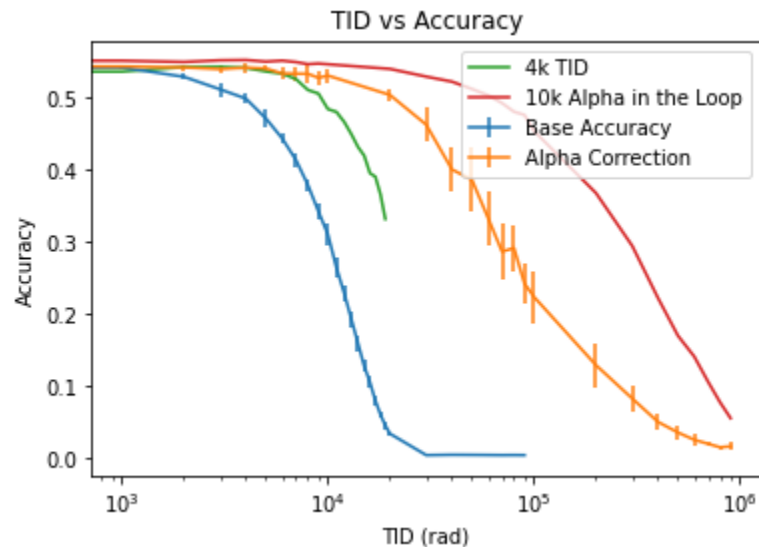


Figure 11: Alpha in the Loop Accuracy

By training the model at 10 krad TID with alpha in the loop, the model was further improved, as evidenced by the 5% roll-off point in the table below:

Training Type	0 TID Accuracy	Peak Accuracy	Loses 5% accuracy TID
Base	54.34	54.34	4.17k
Alpha Correction	54.26	54.38	20.57k
4k Training	54.64	54.64	9.68k
10k Alpha in the Loop	55.2	55.2	57.3k

Table 1: Model 5% Roll-Off Point

Combining both techniques results in a huge increase to 57.3 krad TID when the model loses 5% accuracy, compared to 20.57 krad for just the alpha correction factor and 9.68k for just training with noise. This demonstrates that the techniques used were extremely effective in reducing the effects of radiation on image accuracy on SONOS chips, meeting the target specifications set at the beginning of this report.

6. FUTURE WORK

Although this project successfully demonstrated the effectiveness of the methods of alpha correction factor and training with noise through simulations, it did not include evaluation of these methods under real-life conditions. Real-life assessments are particularly important for the method of alpha correction factor, as it uses column current measurements collected from the board in order to calculate the factor.

Future work may focus on experimentally validating these methods. In particular ASU's radiation chamber can be utilized to perform irradiation tests and collect the data.

7. ABET CRITERION

Sustainability: By implementing machine learning inference upon in-memory analog computing hardware, this will reduce the energy consumption by over 100 times. This is because using an analog crossbar array for matrix multiply only requires one operation compared to the many required from digital. The reduced energy consumption will make machine learning use in the future far more sustainable.

Manufacturability: The SONOS board used in this project is already being mass-produced by Infineon, meaning that in the future it will be quite easy to scale this project to a variety of applications. Instead of using the SONOS board digitally, it simply has to be modified to an analog version, and no new processes need to be created to begin manufacture of the board.

Political: If radiation effects can be fixed, using the SONOS chip for machine learning can then be used in space, which has a variety of political implications. Space exploration has been an area of both collaboration and competition between countries, with the Space Race and ISS being prominent examples. Developing this technology will further push the boundaries of space exploration, which could result in heavy advantages in contentious areas such as ICBMs, space mining, and more.

8. CONCLUSION

Using analog in-memory computing to perform machine learning inference offers a highly energy-efficient solution for environments such as space, where power is limited and reliability is critical. However, radiation-induced conductance drift poses a serious challenge to maintaining accuracy. This project addressed that challenge by implementing a radiation model in a custom Python simulation environment and applying two key techniques: a correction factor to offset drift and noise injection during training to enhance model resilience. These methods were successfully tested in simulation and significantly improved image recognition accuracy on the SONOS chip under high radiation exposure. The results demonstrate that memristive chips, when properly corrected, are a viable and efficient option for machine learning applications in radiation-prone environments.

REFERENCES

- [1] Joshi, V., Le Gallo, M., Haefeli, S. *et al.* Accurate deep neural network inference using computational phase-change memory. *Nat Commun* **11**, 2473 (2020).
<https://doi.org/10.1038/s41467-020-16108-9>

- [2] R. Maurer, M. Fraeman, M. Martin, and D. Roth, "Harsh Environments: Space Radiation Environment, Effects, and Mitigation," *SPACE RADIATION EFFECTS ON ELECTRONICS AND MITIGATION JOhNS hOPKINS APL TEChNICAL DiGEST*, vol. 28, no. 1, 2008, Available:
<https://secwww.jhuapl.edu/techdigest/Content/techdigest/pdf/V28-N01/28-01-Maurer.pdf>

- [3] Siath, M., Xiao, P. *et al.* In-situ analog computing under ionizing radiation with SONOS charge trap memory. (Not published)

- [4] T. Patrick Xiao et al., "In-situ analog in-memory computing under ionizing radiation exposure," in *IEEE Transactions on Nuclear Science*, doi: 10.1109/TNS.2025.3537985.

- [5] "SOURCES AND EFFECTS OF IONIZING RADIATION United Nations Scientific Committee on the Effects of Atomic Radiation UNSCEAR 2000 Report to the General Assembly, with Scientific Annexes VOLUME I: SOURCES UNITED NATIONS." Available:
https://www.unscear.org/docs/publications/2000/UNSCEAR_2000_Report_Vol.I.pdf